

FRONT END DEVELOPMENT FRAMEWORKS AND UI ENGINEERING (25CS1201E)

2025-26, Term-III

Project Problem Statement Submission Form

Sec No-	Team No-	Project Title-Prescription Management System
Team Lead Roll Number		G.S.S Gyanadeep
Team Lead Name		2520080008
Team Member Roll Number		G Mahidhar
Team Member Name		2520090138

1. Abstract

(Summarize the project idea, problem background, and proposed solution. Not more than 250 words.)

The **Prescription Management System** is a web-based healthcare application that digitizes the prescription lifecycle for patients, doctors, and pharmacies. Built using **React.js**, **Node.js**, **Express.js**, and **MySQL**, the system features key modules including a **Prescription Form**, **Drug Search**, **Dosage Validator**, **Refill Request**, **Pharmacy Assignment**, **PDF Export**, **Adherence Tracker**, **Prescription History**, **Notifications**, and an **Admin Panel**. The frontend is developed using **HTML5**, **CSS3**, **JavaScript (ES6+)**, **React Router**, and **Bootstrap/Tailwind CSS**, while **Context API/Redux** manages state and **Axios** handles API communication. The system aims to reduce prescription errors, prevent fraud, and improve overall healthcare delivery efficiency.

2. Module Mapping

(Provide details of how modules are divided among team members.)

Module Name	Assigned Roll Number(s)	Signature

3. Module Details

(Explain the responsibility of each student concerning their assigned module.)

- 1. Prescription Form** Create and submit digital prescriptions with patient, doctor, and medication details.
- 2. Drug Search** Search drugs from a database with name-based suggestions and basic drug info.
- 3. Dosage Validator** Validates prescribed dosage against safe limits to avoid medication errors.
- 4. Refill Request** Patients request prescription refills; doctors approve or reject them digitally.
- 5. Pharmacy Assign** Assigns and routes approved prescriptions to the nearest available pharmacy.
- 6. PDF Export** Downloads a formatted PDF copy of the prescription for records and sharing.
- 7. Adherence Tracker** Tracks whether patients follow their medication schedule with reminders.
- 8. History** Stores and displays complete prescription and medication history for reference.
- 9. Notifications** Sends alerts for medication timings, refill reminders, and prescription updates.
- 10. Admin Panel** Manages users, drug database, pharmacies, and overall system settings.

4. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

Frontend

- **React.js** — UI development with reusable components
- **React Router** — Page navigation and routing
- **Bootstrap / Tailwind CSS** — Responsive styling and design
- **HTML5 & CSS3** — Structure and styling
- **JavaScript (ES6+)** — Core scripting logic

Backend

- **Node.js** — Server-side runtime environment
- **Express.js** — REST API development and routing

Guide Name and Signature (remarks if any) _____